Capstone Design Project — Final Report

The Effect of Reward Function Design in RL-Based BTC Grid Trading: Pareto Frontier, Winner Reversal, and Out-of-Distribution Robustness of Prospect-Theoretic Reward

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Abstract

We quantitatively study the effect of *reward function design* on PPO-based reinforcement-learning policies for ATR-proportional grid trading on the BTC/USDT 1-hour market. Four reward variants—symmetric (`sym`), asymmetric ($\beta = 3.42$, `asym`), Differential Sharpe Ratio ($\eta = 1/28\,\text{h}$, `dsr`; Moody and Saffell 2001), and prospect-theoretic ($\alpha = 0.68$, $\lambda = 3.30$, `pt`; Kahneman and Tversky 1979)—are each trained with 10 seeds × 1M steps and evaluated on three environments: (i) single-split validation (Val 2021–2023), (ii) 6-fold combinatorial purged cross-validation [López de Prado, 2018] 15 paths, and (iii) the unsealed out-of-sample Test set (2024–2026, BTC \$42K → \$76K bull regime).

We report five key findings. **(1)** All four variants exceed the ATR baseline by +11–24% on Val, yet partition into two clusters {`sym`, `dsr`} (aggressive) vs. {`asym`, `pt`} (conservative); within-cluster Cohen's $|d| < 0.3$ versus across-cluster $|d| > 0.8$. **(2)** Mechanism: reward loss asymmetry directly determines trade frequency and holding time (policy distance ratio 2.22×), and the sliding-window memory structure of DSR results in 2–6× higher hold rate than other variants. **(3)** Winner reverses across evaluation environments: Val=`sym`(1.87) → CPCV=`dsr`(1.41, $p < 0.001$) → Test=`pt`(0.34–0.37, consistent across two model sources, $p < 0.002$). **(4)** The OOS robustness of `pt` arises from a short-holding mechanism: loss aversion $\lambda = 3.30$ combined with concave gain $\alpha = 0.68$ trains policies to exit within mean 1.4 h (max 6 h), thereby avoiding the sell-side timing risk of the unseen bull regime. In contrast, `dsr` learns long holding (mean 4.58 h, max 169 h = 7 days) and records the worst OOS performance—demonstrating that the same reward formulation underlies both in-sample advantage and OOS failure as two sides of the same coin. **(5)** The Val–Test distribution shift is highly significant (KS $p < 10^{-10}$, −27% variance), while policies' actions remain nearly invariant between Val and Test ($\Delta \leq 5\%$), indicating that the regime shift, not the policy, drives OOS differences.

Our contribution is twofold. First, we demonstrate quantitatively that reward design exerts its effect through a *trade-off dimension in risk profile* rather than a single Sharpe "alpha source." Second, we provide the first quantitative evidence that Kahneman-Tversky prospect theory confers out-of-distribution safety on RL trading policies. Code and data: https://github.com/cosmicpotato2047/capstone-rl-trading.

Keywords: Reinforcement Learning, Grid Trading, PPO, Reward Design, Prospect Theory, Differential Sharpe Ratio, Combinatorial Purged CV, Out-of-Distribution Generalization, Cryptocurrency

Contents

1 Introduction

Cryptocurrency markets exhibit micro-oscillations in hourly price that are largely orthogonal to long-term trends, motivating *grid trading* strategies that harvest revenue from volatility itself rather than from directional prediction. Grid trading is a simplified form of market making [Avellaneda and Stoikov, 2008], ranging from fixed-grid rules to volatility-adaptive variants that scale grid width by the Average True Range (ATR) [Wilder, 1978]. This work studies the latter: an ATR-proportional grid system whose grid width and limit-order placement are determined by a reinforcement-learning policy.

1.1 Motivation and Research Question

An earlier phase of this project hypothesized that PPO RL would discover dynamic volatility adaptation and outperform a fixed-grid baseline. Post-hoc behavioral analysis, however, revealed that the trained policy converged to a near-constant action $[1.0, 0.0]$ (§4), and that the Bayesian-optimized ATR coefficient itself was the principal source of performance. This negative finding suggested that the symmetric-reward + ATR-proportional combination structurally absorbs the policy's degrees of freedom, motivating the new research question of this paper:

> **In BTC grid trading, how does the design of the reward function affect the behavioral pattern and generalization performance of RL policies? In particular, do loss-asymmetric (asymmetric, prospect-theoretic) or risk-adjusted-return (DSR) reward formulations yield meaningful differences in single-metric advantage or out-of-sample robustness?**

1.2 Contributions

This work makes the following four contributions.

1. **A statistically rigorous comparison framework for reward variants.** Four reward functions (`sym`, `asym`, `dsr`, `pt`) are trained in an identical environment with 10 seeds × 1M steps, and pairwise statistical tests (Cohen's d, bootstrap P(A>B), IQM, 5% CVaR) are applied following Henderson et al. [2018]'s reproducibility guidelines.

2. **Scenario D — Pareto-frontier discovery.** Rather than a single "best variant" winner, the four variants statistically partition into two clusters {`sym`, `dsr`} vs. {`asym`, `pt`} forming a Pareto-like frontier on the Sharpe–MDD plane (§5). This outcome falls into none of the pre-registered scenario branches (A optimistic / B neutral / C pessimistic).

3. **Causal chain: reward → behavior → outcome.** Policy-level differences across variants are quantified on 1.04M step trajectories via a policy-distance matrix (within-cluster 0.129 vs. across-cluster 0.286), regime- specific trade/hold statistics, and action distribution comparison, confirming the mechanism reward loss asymmetry → reduced trade frequency → conservative cluster (§6).

4. **Winner reversal across evaluation environments and OOS robustness of prospect theory.** The winner reverses completely across single-split Val (`sym` 1$^{\text{st}}$) → CPCV multi-split

(`dsr` 1st) → the unsealed Test (`pt` 1st, consistent across two sources $p < 0.002$) (§7). The OOS robustness of `pt` is mechanistically attributed via trajectory analysis to its loss aversion $\lambda = 3.30$ and concave utility $\alpha = 0.68$ training policies to exit within mean 1.4 h, thereby avoiding the sell-side timing risk of the unseen bull regime. This is, to our knowledge, the first quantitative demonstration that the Kahneman-Tversky prospect theory [Kahneman and Tversky, 1979] confers OOS safety on RL trading policies.

1.3 Paper Structure

Section 2 surveys related work on RL trading, reward design theory, and evaluation methodology. Section 3 details the environment design (MDP, ATR formula, four reward variants) and PPO training. Section 4 briefly reproduces the Phase 1 negative finding ("RL = Fixed $[1.0, 0.0]$") as motivation. Section 5 reports the two-cluster Pareto-frontier result on Val (Scenario D), while Section 6 quantifies the mechanism. Section 7 presents the robustness trio: slippage (§7.1), CPCV multi-split (§7.2), and the central contribution of this paper—the Test OOS evaluation and the mechanistic basis of prospect theory's robustness (§7.3). Section 8 discusses distribution shift quantification and limitations, and Section 9 concludes.

2 Related Work

This paper sits at the intersection of four areas: (i) market making and grid trading, (ii) reinforcement-learning-based trading, (iii) reward-design theory, and (iv) evaluation methodology for trading policies. This section surveys the key prior work in each area and clarifies where our contribution lies.

2.1 Market Making and Grid Trading

The academic ancestor of grid trading is the market-making model of Avellaneda and Stoikov [2008]. Their analysis derives a closed-form optimal policy for a risk-averse market maker who faces inventory risk and must place *asymmetric* bid/ask quotes around the mid-price. Two key insights are that (a) the bid–ask spread is a function of volatility and remaining time horizon, and (b) inventory skew shifts the quote midpoint asymmetrically. Grid trading is a simplified practical variant of this model in which orders are placed on a pre-defined ladder of price levels in order to monetize micro-volatility.

The standard mechanism for adapting grid widths to market volatility is the Average True Range [Wilder, 1978] (ATR; 14-period moving average is the default). ATR is a smoothed average of the true price range and provides a simple, robust rule that automatically widens the grid in volatility-cluster regimes. ATR-proportional grids of the form $\Delta = c \cdot \mathrm{ATR}$ (with proportional coefficient c) are widely used in practice and serve as the baseline of this paper.

Most prior academic work in this area is limited to the analysis of *rule-based* policies, with relatively few attempts to learn the grid-width decision from data. This paper adopts a structure in which the proportional coefficient and the number of grid splits in the ATR formula are dynamically determined by an RL policy, while quantitatively analyzing how the design of the reward function influences the resulting learned policy's behavior—a combination, to our knowledge, that has not been explored.

2.2 Reinforcement Learning for Trading

The standard DRL baseline for cryptocurrency trading is the PPO/A2C/DQN comparison framework of Zhang et al. [2020], which feeds price time series directly into a PPO policy and shows consistent improvements over simple buy-and-hold (B&H) and moving-average crossover strategies. The ACM survey of Sun et al. [2023] and the more recent review of Liu [2025] both identify two open problems in the area: *(a) interpretability of policy behavior and (b) generalization across market regimes*. This paper addresses both directly: §6 quantifies the causal chain from reward form to behavioral difference, and §7 documents out-of-sample (OOS) generalization phenomena that are not captured by single-split validation alone.

There are four directly preceding works. Liu et al. [2020] feeds the output of an LSTM price predictor into the state of a PPO trading policy in a two-stage pipeline. The limitation of this approach is that prediction errors propagate directly into trading decisions; this paper bypasses the limitation by *eliminating* the price-prediction stage and using a volatility-adaptive grid structure instead. Yasin and Gill [2024] compares DQN/PPO/A2C with 20 technical indicators, but the action space is discrete buy/sell with no hold action. This paper adopts a continuous two-dimensional action (aggressiveness and profit target) that supports fine-grained grid control. Pham et al. [2025] uses DQN to select one of five predefined strategies as a meta-strategy, but the strategies themselves are not learned. Bandarupalli [2025] augments PPO with a volatility penalty and a transaction-cost term, but stays at the level of hyperparameter tuning within a single reward variant and does not compare across reward *forms*, which is the focus of this paper.

A particularly relevant prior work is Gort et al. [2022], which empirically shows that *single-split train/test results do not guarantee OOS generalization* in cryptocurrency DRL trading, and recommends CPCV (Combinatorial Purged Cross-Validation) and DSR (Deflated Sharpe Ratio). This paper adopts those recommendations, using Val (2021–2023), CPCV (6-fold inside the Train partition), and Test (2024– unsealed) environments concurrently. Our *winner reversal* finding (Val `sym` $\rightarrow$ CPCV `dsr` $\rightarrow$ Test `pt`) is a quantitative confirmation that Gort et al.'s concern reappears in the dimension of reward-variant selection.

In one sentence, our position is: whereas prior DRL trading literature focuses on comparing algorithms and state designs under a *single* reward function, this paper performs a controlled comparison of *four reward functions* on top of the same algorithm (PPO) and the same environment, and separately measures their effects on in-sample / multi-split / OOS environments.

2.3 Reward-Design Theory

The four reward variants compared in this paper draw from three theoretical lineages in reinforcement learning and behavioral economics. (i) Symmetric (`sym`) is the standard PnL-return reward and serves as our baseline. (ii) Asymmetric (`asym`) places a positive weight $\beta > 1$ on losses, a simple first-order approximation of a risk-averse utility. (iii) Differential Sharpe Ratio (`dsr`) is the formulation of Moody and Saffell [2001]. (iv) Prospect-theoretic reward (`pt`) directly uses the utility function of Kahneman and Tversky [1979] and Tversky and Kahneman [1992], $v(x) = \mathrm{sgn}(x)|x|^\alpha$ for $x \geq 0$ and $-\lambda|x|^\alpha$ for $x < 0$, as a per-step reward.

The DSR formulation of Moody and Saffell [2001] uses the *differential change of the online Sharpe-ratio estimator* as the step reward, tracking first and second moments via exponentially-weighted moving averages (EWMA, decay rate η). This has two implications for RL training: (a) it directly optimizes a risk-adjusted return rather than raw PnL, and (b) the sliding-window memory imposes a *multi-step temporal dependency* on the policy. As we show in §6, this memory structure is what causes the `dsr` policy to learn 2–6× longer holding times than the other variants—a property that becomes the in-sample advantage of §7 and the OOS failure of §7.3 as two sides of the same causal coin.

Prospect theory [Kahneman and Tversky, 1979] is a foundational model of human decision making under risk, with two characteristic parameters: (a) loss aversion $\lambda \approx 2.25$ (standard value from Tversky and Kahneman 1992) and (b) the concavity exponent $\alpha \approx 0.88$. While the theory has been widely used to describe human trader behavior (e.g., the disposition effect of clinging to losses and realizing gains too early), *direct use of prospect-theoretic utility as an RL reward* is rare in the literature. To our knowledge, this paper is the first to directly implement the Kahneman and Tversky [1979] utility as an RL reward (with $\alpha = 0.68$, $\lambda = 3.30$, tuned by Optuna) and quantitatively establish the OOS robustness that the resulting policy exhibits.

Ng et al. [1999]'s policy-invariance theorem of potential-based reward shaping states that "a monotone transformation of the reward signal does not alter the policy learned by RL." At first glance, this paper's comparison of four reward variants may appear to conflict with the theorem, but it does not: the `sym` → `asym`/`pt` transformations are *non-potential-based* (they apply an asymmetric scale on the negative half-line), while `dsr` is not a simple per-step reward transformation but a *state-extended* reward (the sliding-window moments effectively become part of the state). Our reward-variant comparison thus lies outside the scope of the Ng et al. theorem, and indeed we confirm in §6 that policy behavior differs across variants in a statistically significant manner.

2.4 Evaluation Methodology

The two principal pitfalls in evaluating RL trading policies are (a) backtest overfitting and (b) seed dependence of results. Henderson et al. [2018] empirically demonstrates that seed variability alone can produce large differences for the same algorithm, and proposes 5–10 seed statistical comparisons, IQM (Interquartile Mean), and bootstrap confidence intervals as standard recommendations. This paper uses 10 seeds per variant (Val/exp033) and 100 models (Test), and applies pairwise Cohen's d and bootstrap $\mathrm{P}(A > B)$ to all comparisons.

López de Prado [2018] (Ch. 8) proposes *Combinatorial Purged Cross-Validation* (CPCV), which avoids train/test leakage in time-series data while leveraging $\binom{N}{k}$ split combinations to strengthen the generalization guarantees of single-split results. We adopt 6-fold CPCV (15 paths) in §7.2.

Bailey and López de Prado [2014]'s *Deflated Sharpe Ratio* (DSR*) is a statistical-significance test on the Sharpe ratio that corrects for the number of backtest trials and for the skewness/kurtosis of the underlying distribution. We apply Bonferroni 4-way correction together with DSR* testing to the multiple comparisons of CPCV results (§7.2), and confirm that all four reward variants are significant relative to the ATR baseline at $p < 0.001$.

The methodological distinctiveness of this paper is the *simultaneous use of three evaluation environments* (Val / CPCV / Test) that allowed us to discover the winner reversal. Studies relying solely on single-split or solely on CPCV cannot expose OOS generalization failure, and studies relying solely on OOS cannot expose the in-sample conservatism of `pt`. The simultaneous application of three environments is, in our view, a methodological contribution: it generalizes the multi-evaluation recommendations of Henderson et al. [2018] and Gort et al. [2022] to the single dimension of reward-variant comparison.

3 Method

3.1 MDP Formulation

We cast single-asset dynamic grid trading on the BTC/USDT 1-hour series as a finite-horizon Markov decision process $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, R, \gamma)$. An episode runs for 720 hours (30 days) starting from a randomly chosen step within the training/validation range, and each step t corresponds to a 1-hour price bar.

State $\mathcal{S} \subset \mathbb{R}^7$. The observation $\mathbf{s}_t = (s_t^{(0)}, \ldots, s_t^{(6)})$ couples market and portfolio state:

$$s_t^{(0)} = \log\Big(p_t \,/\, \overline{p}_{t,168}\Big) \qquad \text{(price vs. 1-week mean)} \tag{1}$$

$$s_t^{(1)} = \frac{\bar{p}_t^{\text{avg}} - p_t}{\bar{p}_t^{\text{avg}}} \qquad \text{(divergence from average cost)} \tag{2}$$

$$s_t^{(2)} = \frac{h_t \cdot p_t}{C_0} \qquad \text{(holdings value / capital)} \tag{3}$$

$$s_t^{(3)} = \frac{c_t}{C_0} \qquad \text{(cash ratio)} \tag{4}$$

$$s_t^{(4)} = \frac{\text{ATR}_{168}(p)}{p_t} \qquad \text{(volatility)} \tag{5}$$

$$s_t^{(5)} = \frac{p_t - p_{t-72}}{p_{t-72}} \qquad \text{(short-term trend, 3d)} \tag{6}$$

$$s_t^{(6)} = \frac{p_t - p_{t-720}}{p_{t-720}} \qquad \text{(long-term trend, 30d)} \tag{7}$$

Here p_t is the bar close, $\bar{p}_t^{\text{avg}}$ is the average cost of the current position (set to 0 when no position is held, in which case $s_t^{(1)}$ is also 0), h_t is the BTC quantity, c_t the cash balance, and $C_0 = 10{,}000$ USDT is the initial capital. ATR_{168} is the Average True Range [Wilder, 1978] computed over a 168-bar (1-week) window, and $\overline{p}_{t,168}$ is the corresponding simple moving average. All components are pre-normalized by a 168-bar rolling z-score computed on the training set and clipped to

$[-5, 5]$.

Action $\mathcal{A} = [0,1]^2$. At each step the policy emits a continuous 2D action $\mathbf{a}_t = (a_t^{(0)}, a_t^{(1)})$. The first coordinate $a_t^{(0)}$ (*aggressiveness*) controls two buy limit orders, and $a_t^{(1)}$ (*profit target*) controls two sell limit orders. We avoid a discrete buy/sell/hold action space because fine-grained grid width adjustment and volatility adaptation are most naturally expressed in continuous values; discretization would artificially limit the expressiveness of the action space.

ATR-proportional grid formula. Grid widths and positions are determined through volatility-normalized action coordinates. With $r_t = \mathrm{ATR}_{168}(p)/p_t$, current price p_t, and average cost $\bar{p}_t^{\mathrm{avg}}$,

$$g_t^{\mathrm{buy\_hi}} = r_t \cdot (A_b + B_b \cdot a_t^{(0)}) \tag{8}$$

$$g_t^{\mathrm{buy\_lo}} = r_t \cdot (C_b + D_b \cdot a_t^{(0)}) \tag{9}$$

$$g_t^{\mathrm{sell\_mkt}} = r_t \cdot (A_s + B_s \cdot a_t^{(1)}) \tag{10}$$

$$g_t^{\mathrm{sell\_cost}} = r_t \cdot (C_s + D_s \cdot a_t^{(1)}) \tag{11}$$

Quote prices:

$$q_t^{\mathrm{buy\_hi}} = p_t(1 - g_t^{\mathrm{buy\_hi}}), \quad q_t^{\mathrm{buy\_lo}} = p_t(1 - g_t^{\mathrm{buy\_lo}}) \tag{12}$$

$$q_t^{\mathrm{sell\_mkt}} = p_t(1 + g_t^{\mathrm{sell\_mkt}}), \quad q_t^{\mathrm{sell\_cost}} = \bar{p}_t^{\mathrm{avg}}(1 + g_t^{\mathrm{sell\_cost}}) \tag{13}$$

The coefficients $\{A_b, B_b, C_b, D_b, A_s, B_s, C_s, D_s\}$ are fixed environment constants, with values reported in §3.3. Only the `sell_cost` quote is anchored at the average cost; the choice explicitly exposes "break-even recovery" to the policy.

Reward. The baseline (`sym`, symmetric) form is the 1-step PnL ratio:

$$r_t^{\mathrm{sym}} = \frac{E_t - E_{t-1}}{C_0}, \quad E_t = c_t + h_t \cdot p_t \tag{14}$$

Transaction fees are already deducted from c_t inside `_execute_buy` and `_execute_sell`, so no separate subtraction appears in the reward. All four reward variants in this paper are defined as functions of r_t^{sym} and detailed in §3.4.

3.2 Trading-Environment Dynamics

This subsection details order execution, the cycle definition, and the transaction-cost model.

Limit-order execution. Fill decisions for the four quotes use the next bar's high/low $(p_{t+1}^{\mathrm{H}}, p_{t+1}^{\mathrm{L}})$:

$$\begin{aligned}
p_{t+1}^{\mathrm{L}} \le q_t^{\mathrm{buy_hi}} &\Rightarrow \text{fill at } q_t^{\mathrm{buy_hi}} \\
p_{t+1}^{\mathrm{L}} \le q_t^{\mathrm{buy_lo}} &\Rightarrow \text{fill at } q_t^{\mathrm{buy_lo}} \\
p_{t+1}^{\mathrm{H}} \ge q_t^{\mathrm{sell_mkt}} &\Rightarrow \text{fill at } q_t^{\mathrm{sell_mkt}} \\
p_{t+1}^{\mathrm{H}} \ge q_t^{\mathrm{sell_cost}} &\Rightarrow \text{fill at } q_t^{\mathrm{sell_cost}}
\end{aligned} \tag{15}$$

When both buy and sell quotes are filled in the same bar we apply a *sell-first* rule, prioritizing inventory liquidation. The fill price is the *quote price itself*, not the next-bar extreme. Filling at the next-bar extreme injects favorable bias into the simulation; we therefore discarded an earlier "next_low/next_high" execution rule and use only the limit-order rule throughout this paper.

Cycle definition. A cycle starts when the *first buy fills while holdings = 0* and ends when subsequent sells return holdings to 0. Multiple partial buys/sells can occur inside a single cycle, and the cycle return is $\text{cycle\_pnl} = (c_{\text{end}} - c_{\text{start}})/c_{\text{start}}$. At the start of each cycle, the slot size used for the cycle is set:

$$\text{slot\_size}_{\text{cycle}} = \frac{c_{\text{start}}}{n_{\text{splits}}}, \quad \text{order\_size} = \frac{\text{slot\_size}}{n_{\text{buy\_orders}}} \tag{16}$$

We use $n_{\text{splits}} = 2$ and $n_{\text{buy\_orders}} = 2$ (the best values from Phase 2 Optuna Trial #34). `threshold_basis` is fixed to `avg_price`, meaning the "liquidate all" detection uses the average cost.

Transaction-cost model. Transaction costs are set to *0.05% per side* (Binance maker fee). On a buy, the filled quantity is $\Delta h = \text{order\_size} \cdot (1 - f)/q$ and the cash change is $\Delta c = -\text{order\_size}$, so the fee is already reflected in c_t. In §7.1 (exp033) we add an additional slippage $\sigma = 0.02\%$ that modifies fill prices to $q^{\text{buy}} \cdot (1 + \sigma)$ and $q^{\text{sell}} \cdot (1 - \sigma)$. Slippage is not applied to the base environment and appears only in the robustness analysis.

3.3 ATR Baseline

The comparison baseline *shares the environment and grid formula* with our RL policies but determines actions via a rule. Concretely, the ATR baseline reuses the grid formulas of §3.1 for $g^{\text{buy\_hi}}, g^{\text{buy\_lo}}, g^{\text{sell\_mkt}}, g^{\text{sell\_cost}}$ with all RL action degrees of freedom *zeroed* ($a_t^{(0)} = a_t^{(1)} = 0$, so that the B_b, D_b, B_s, D_s terms drop and the grid widths are fixed time-independent constants determined entirely by A_b, C_b, A_s, C_s). The coefficient values come from Phase 2 Optuna Trial #34:

Table 1: Grid coefficients of the ATR baseline (Phase 2, Optuna Trial #34).

Coef.	Value	Role	Coef.	Value	Role
A_b	1.665	buy_hi base width	A_s	0.285	sell_mkt base width
C_b	6.070	buy_lo base width	C_s	1.951	sell_cost base width
B_b	1.748	buy_hi action sensitivity	B_s	1.950	sell_mkt action sens.
D_b	18.683	buy_lo action sensitivity	D_s	7.500	sell_cost action sens.

The ATR baseline yields Val Sharpe 1.378 without slippage,[1] dropping to 0.835 when slippage of 0.02% is added.

[1]In the early RESEARCH_LOG (May 14, 2026) we recorded 1.505; the value was re-measured to 1.378 after the Phase 16a unification of the evaluation setup ($n_{\text{eval\_episodes}}$, env config synchronization). We use 1.378 throughout the paper.

3.4 Four Reward Variants

We now define the four reward variants that form the principal comparison unit of this paper. All are expressed as functions of $x_t := r_t^{\mathrm{sym}} = (E_t - E_{t-1})/C_0$ and differ *only* in functional form; the state, action, ATR grid, and fee model are otherwise identical.

(i) Symmetric (`sym`).

$$r_t^{\mathrm{sym}} = x_t = \frac{E_t - E_{t-1}}{C_0} \tag{17}$$

No hyperparameter. Baseline of this paper.

(ii) Asymmetric (`asym`).

$$r_t^{\mathrm{asym}} = \begin{cases} x_t & \text{if } x_t \geq 0 \\ \beta \cdot x_t & \text{if } x_t < 0 \end{cases}, \quad \beta > 1 \tag{18}$$

A first-order approximation of a risk-averse utility, weighting losses by β. An Optuna 30-trial $\times$ 200k-step search over the range $[1.0, 4.0]$ to maximize Val Sharpe selected the best $\beta = 3.420$.

(iii) Differential Sharpe Ratio (`dsr`). The formulation of Moody and Saffell [2001]. Given an EWMA decay rate η, online estimates of the first and second return moments evolve as

$$A_t = A_{t-1} + \eta\,(x_t - A_{t-1}) \tag{19}$$

$$B_t = B_{t-1} + \eta\,(x_t^2 - B_{t-1}) \tag{20}$$

$$\Delta A_t = x_t - A_{t-1}, \quad \Delta B_t = x_t^2 - B_{t-1} \tag{21}$$

The DSR step reward is the differential of the on-line Sharpe estimator $\hat{S}_t = A_t/\sqrt{B_t - A_t^2}$:

$$r_t^{\mathrm{dsr}} = \frac{B_{t-1}\,\Delta A_t - \frac{1}{2}A_{t-1}\,\Delta B_t}{(B_{t-1} - A_{t-1}^2)^{3/2}} \tag{22}$$

A 30-trial Optuna search over $[1/720, 1/24]$ (log-uniform) selected the best $\eta = 0.0352 \approx 1/28\,\mathrm{h}$, an EWMA window of approximately 1.2 days.

(iv) Prospect-theoretic (`pt`). The utility function of Kahneman and Tversky [1979] and Tversky and Kahneman [1992] applied directly as a step reward:

$$r_t^{\mathrm{pt}} = \begin{cases} \mathrm{sgn}(x_t)\,|x_t|^{\alpha} & \text{if } x_t \geq 0 \\ -\lambda\,|x_t|^{\alpha} & \text{if } x_t < 0 \end{cases}, \quad 0 < \alpha \leq 1, \quad \lambda \geq 1 \tag{23}$$

$\alpha < 1$ encodes the concavity of the utility (diminishing marginal utility of gains), and $\lambda > 1$ encodes loss aversion. The standard values from Tversky and Kahneman [1992], fitted to human behavior, are $\alpha = 0.88$ and $\lambda = 2.25$. The Val-Sharpe-maximizing values for our RL policy, however, are $\alpha = 0.683$ and $\lambda = 3.303$, found by Optuna 30 trials over $\alpha \in [0.5, 1.0]$ and $\lambda \in [1.0, 4.0]$. The fact that RL prefers a *more concave* utility and *stronger loss aversion* than the human standard is consistent with the mechanism in §6 and the OOS robustness reported in §7.3.

Optuna search summary. For each of the three variants `asym`, `dsr`, `pt` we ran 30 trials × 200k steps using a TPE sampler [Akiba et al., 2019] (seed=42, MedianPruner with $n_{\text{startup}} = 5$). The total training cost was $3 \times 30 \times 200\text{k} = 18\text{M}$ steps, taking approximately 2 hours on a Windows 11 CPU with $n_{\text{envs}} = 4$. `sym` has no hyperparameter and skips this stage.

3.5 PPO Training and the Stabilization Package

Policy training uses PPO [Schulman et al., 2017] from stable-baselines3 [Raffin et al., 2021]. Standard PPO hyperparameters ($\gamma = 0.99$, $\lambda_{\text{GAE}} = 0.95$, clip $\epsilon = 0.2$, v_f coef $= 0.5$, gradient clip $= 0.5$, MLP policy $[64, 64]$, batch=256, $n_{\text{epochs}} = 10$, $n_{\text{steps}} = 4096$, $n_{\text{envs}} = 4$) follow stable-baselines3 defaults. To mitigate the reward sparsity of our environment (the per-step reward is near 0 when no fill occurs) and the late-stage policy collapse observed in Phase 2 (exp020–022), we add the following *stabilization package* (exp030):

1. **Learning-rate linear decay**: α_t linearly decays from 3×10^{-4} to 1×10^{-5} over training. This bounds the size of late-stage policy updates and dampens post-best collapse.
2. **Entropy coefficient annealing**: c_{ent} linearly anneals from 0.01 to 0.001, ensuring sufficient initial exploration while encouraging convergence to a near-deterministic policy late in training.
3. **Target KL early stop**: target_kl = 0.02 measures the KL divergence after each minibatch update and aborts the update for that batch if the threshold is exceeded, preventing PPO from making myopic large policy changes.
4. **Best-model checkpoint**: Val Sharpe is evaluated every 50k steps ($n_{\text{eval\_episodes}} = 5$) and the weights at the best Val Sharpe are saved separately, providing protection against late-stage collapse.

The package was developed in exp030 (with `sym` reward) and produced Val Sharpe best 1.974 at step 550k, falling to final 1.209 at step 1M. Because the collapse pattern after step 700k persists despite the package, *all evaluations in this paper use the best checkpoint*; the final model is never used.

3.6 Evaluation Protocol

We use *three evaluation environments concurrently.* The data partition is strict time-order to prevent leakage:

Table 2: Data partition.

Partition	Range	Bars (approx.)
Train	2017-08-17 – 2020-12-31	∼30,000
Val (single-split)	2021-01-01 – 2023-12-31	∼26,000
Test (sealed)	2024-01-01 – 2026-04	∼20,000

Val (single-split). The main comparison environment of §5. For each reward variant we train 10 seeds × 1M steps and evaluate the best_model on the entire Val range with $n_{\text{eval\_episodes}} = 5$, producing per-seed Sharpe/MDD/Calmar/trade-count/return.

CPCV 6-fold. The multi-split environment of §7.2. Following López de Prado [2018], we apply 6-fold Combinatorial Purged Cross-Validation *inside the Train partition*, producing $\binom{6}{2} =$ 15 paths (each consisting of 5 months of training plus 1 month of evaluation). A 1-week purge gap between train and evaluation folds prevents information leakage. We execute 4 variants × 15 paths × 1 seed = 60 runs.

Test (unsealed). The OOS environment of §7.3. For honesty, following Gort et al. [2022], Test was *kept sealed from any search or hyperparameter decision* until exp035 (May 16, 2026). After unsealing, we evaluate 100 RL models (4 variants × multi-seed/seed-pair) along with the ATR baseline and Buy-and-Hold on Test.

Statistical tests. All pairwise comparisons report Cohen's d and bootstrap $\text{P}(A > B)$ (10^4 resamples). Single-sided t-tests against ATR on CPCV use Bonferroni 4-way correction along with the Deflated Sharpe Ratio test of Bailey and López de Prado [2014]. To report evaluation variability and distributional skew, we follow Henderson et al. [2018] and add IQM (Interquartile Mean) and 5% CVaR as secondary metrics.

4 Phase 1: Policy Saturation under Symmetric Reward

This section briefly reproduces the negative finding of an *earlier* Phase 1 (April 2026) experiment series exp020–022 that precedes this paper. The finding is the starting point that motivates the reward-formulation comparison of §3.4. The Phase 1 experiments were run on *Env-v2* (2D ATR-proportional formula, next-bar-extreme execution) rather than on this paper's canonical Env-v4, so the absolute Sharpe values include a favorable-bias artifact. However, the qualitative observation at the heart of this section—that the *policy converges to a constant*—is environment-independent and remains valid as motivation.

4.1 The Saturation Phenomenon

The Phase 1 setup is almost identical to §3 of this paper: state 7D, action 2D, ATR-proportional grid formula, symmetric reward $r_t = (E_t - E_{t-1})/C_0$. The two differences are (i) execution uses the next-bar extremes rather than limit fills, and (ii) the stabilization package of §3.5 had not yet been introduced.

Three experiments converged to the same conclusion.

(a) exp020 (budget_fraction). $a_t^{(0)}$ was redefined from "aggressiveness" to "fraction of cash budget": at the start of each cycle, the RL policy decides how much of the cash balance to commit to the cycle (0 = no entry, 1 = full commitment). The hypothesis was "commit less in bear regimes and more in bull regimes," but the learned policy converged to $a_t^{(0)} \approx 1.0$ across all regimes.

(b) exp021 (entry_gate). $a_t^{(0)}$ was redefined as a "cycle entry gate" (0 = no entry, 1 = allow entry). The hypothesis was "block entries in bear regimes," but the learned entry rate in the `bear` regime was 99.7%, effectively keeping the gate open at every bar.

(c) exp022 (aggressiveness, identical semantics to this paper). $a_t^{(0)}$ was restored to "aggressiveness" and PPO was trained for 1M steps. Behavioral statistics of the deterministic policy: across all regimes $a_t^{(0)} \approx 0$ and $a_t^{(1)} \approx 0$ (both medians equal to 0.0000, means ≤ 0.001). The most decisive diagnostic is that the *raw network outputs* were $[-9.19, -4.30]$, deep in the saturation region of the squashing nonlinearity.

4.2 Decisive Ablation: "RL = Fixed $[1.0, 0.0]$"

We took the converged behavioral statistics and ran an ablation that replaces the RL policy with a *fixed* action. In the `aggressiveness` semantics, $a_t = [1.0, 0.0]$ corresponds to the learned policy's behavior ($a_t^{(0)} \to 1$ produces the narrowest buy spacing in the ATR grid formula and $a_t^{(1)} \to 0$ produces the narrowest sell spacing).

Table 3: Phase 1 (Env-v2) decisive ablation. Absolute Sharpe values reflect the next-bar-extreme execution favorable bias, but the *agreement* between RL and Fixed $[1.0, 0.0]$ is environment-independent.

Policy	Val Sharpe	Test Sharpe
RL exp020 (`sym` reward, 1M steps)	**45.390**	42.090
Fixed $[1.0, 0.0]$ (RL convergence point)	**45.390**	41.769
Fixed $[1.0, 0.5]$	4.116	4.683
Fixed $[1.0, 1.0]$	1.060	1.843

The decisive observation in Table 3 is that RL and Fixed $[1.0, 0.0]$ agree on Val Sharpe *to three decimal places*, meaning the result of 1M steps of PPO training reduces to a constant policy. The remaining Fixed policies $[1.0, 0.5]$ and $[1.0, 1.0]$ are substantially worse, partly justifying the claim that "RL did pick a meaningful point in the action space." The crucial qualifier is that the point is time-independent.

4.3 Mechanism: The ATR Formula Absorbs Degrees of Freedom

The structural cause of policy saturation lies in the grid formula itself. Recalling from §3.1, the sell-side grid width is

$$g_t^{\text{sell\_mkt}} = \frac{\text{ATR}_{168}(p)}{p_t} \cdot (A_s + B_s \cdot a_t^{(1)}) \tag{24}$$

and the term ATR_{168}/p_t *already* reflects market volatility. In a bear regime (ATR↑) the grid widens automatically, producing a conservative position; in a bull regime (ATR↓) the grid narrows, producing an aggressive position. The role left for RL is "how should I further adjust the width that ATR already chose?"—and under a symmetric reward the only optimization direction is *maximize cycle count* $\rightarrow$ *maximize compounding*, whose unique optimum is to pick the narrowest buy spacing ($a_t^{(0)} \rightarrow 1$) and the narrowest sell spacing ($a_t^{(1)} \rightarrow 0$). There is no incentive to learn a state-dependent action.

4.4 Motivation for This Paper

Read literally, the Phase 1 negative finding suggests "RL adds no value to BTC grid trading." This paper takes a different stance: that the conclusion is conditional on the specific assumption of a *symmetric reward*. That is:

> *If the combination of symmetric reward and ATR-proportional formula absorbs the RL degrees of freedom, can reformulating the reward as* asymmetric *(asymmetric, prospect-theoretic) or* path-dependent *(sliding-window DSR) restore the value-add of RL?*

This is precisely the hypothesis formalized in this paper's main comparison (§3.4, §5). We will further see in §5 that, on the canonical environment Env-v4 (limit-order execution plus the stabilization package), even the symmetric reward yields a policy that exceeds the ATR baseline (Val Sharpe sym = 1.87 > ATR 1.378). The strong Phase 1 negative "RL = Fixed" is therefore a joint result of favorable-bias execution and insufficient training stabilization; once the environment is normalized and training is stabilized, the reward form itself becomes the principal channel through which RL adds value.

5 Pareto Frontier and Scenario D

[TBD. exp032b results: 4 variants × 10 seeds, pairwise statistics, Pareto scatter, two-cluster discovery, hypothesis H1–H3 evaluation, Scenario D confirmation.]

6 Mechanism Quantification

[TBD. exp032c: policy distance matrix, behavior per regime, counterfactual action map, action distribution; causal chain reward → trade frequency → cluster.]

7 Robustness Analysis

7.1 Slippage Robustness

[TBD. exp033 results with slippage 0.02%. Cluster preservation ratio 2.19×; ATR-with-slippage fair-comparison results (Phase 16a).]

7.2 Multi-Split Evaluation via CPCV

[TBD. exp034: 6-fold CPCV 15 paths. DSR $p < 0.001$ (Bonferroni 4-way correction); `dsr` reversal to 1st (1.413, 5% CVaR 0.890); cluster preservation ratio 3.55×.]

7.3 Out-of-Sample Evaluation and Prospect Theory's Robustness

[TBD. Central chapter. exp035: 100 RL models + ATR baseline + Buy-and-Hold on Test. `pt` *1st on Test (both sources $p < 0.002$); `dsr` last on Test. Phase 16d mechanism: hold-duration distribution (DSR 7-day max vs. pt 6h max); causal account of* `pt`*'s OOS robustness.]*

8 Discussion

[TBD. Subsections: §8.1 Val–Test distribution-shift quantification (KS $p < 10^{-10}$, -27% variance); §8.2 honest acknowledgment of environment dependence in exp027_rl prior evidence (Env-v3 `asym` *1.955); §8.3 limits of single-metric winner; §8.4 limitations: BTC-only, single regime in Test, single slippage level.]*

9 Conclusion

[TBD. Four main messages: (1) Pareto-frontier trade-off; (2) winner reversal across evaluation environments; (3) OOS robustness of Prospect Theory; (4) in-sample advantage $\neq$ OOS robustness. Future work: multi-asset, additional regimes, DR, meta-RL.]

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